

EXP 10 -RECOMMENDATION SYSTEMS USING MATRIX FACTORIZATION (SVD OR NMF)

1.SCENARIO 1:MATRIX FACTORIZATION USING SVD

OUTPUT:

JANASREE 24BAD040				
userId	movieId	rating		title
0	1	31	2.5	Dangerous Minds (1995)
1	1	1029	3.0	Dumbo (1941)
2	1	1061	3.0	Sleepers (1996)
3	1	1129	2.0	Escape from New York (1981)
4	1	1172	4.0	Cinema Paradiso (Nuovo cinema Paradiso) (1989)

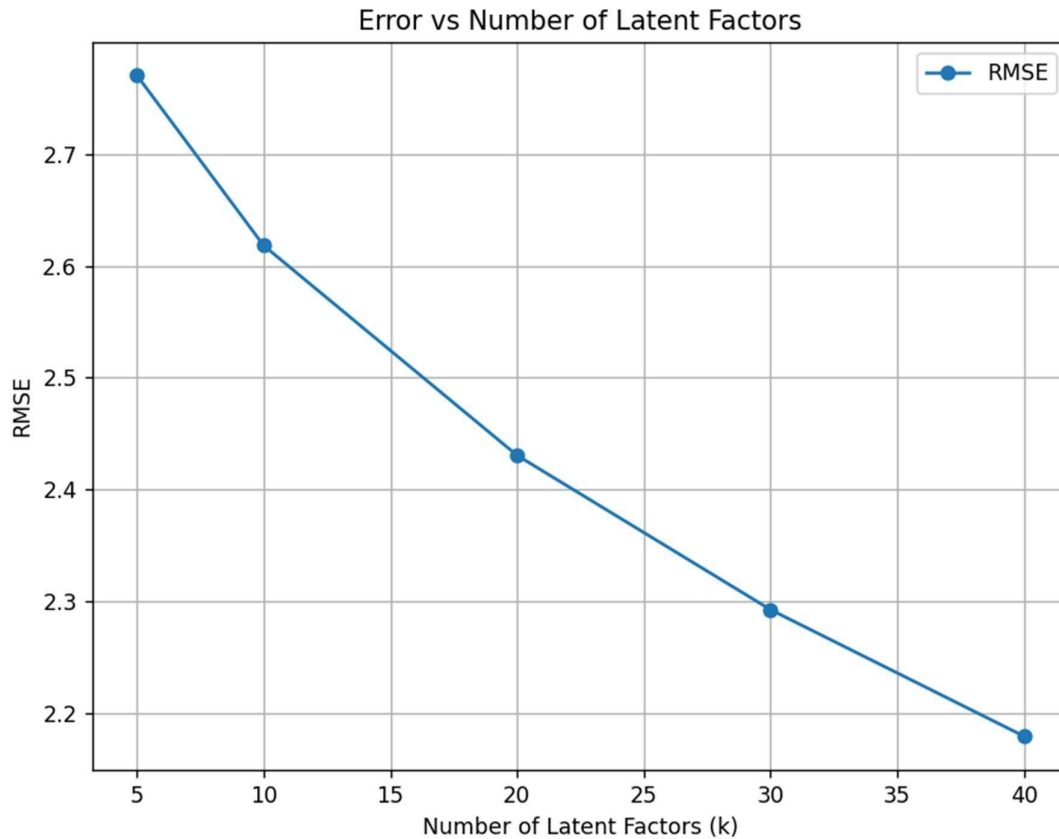
Top 10 Recommended Movies for User 1

title	
Alien (1979)	0.175545
Star Trek II: The Wrath of Khan (1982)	0.163278
Fantasia (1940)	0.159034
Jaws (1975)	0.153054
Monty Python and the Holy Grail (1975)	0.146222
Lawrence of Arabia (1962)	0.144503
African Queen, The (1951)	0.142674
Monty Python's Life of Brian (1979)	0.142128
Who Framed Roger Rabbit? (1988)	0.136410
Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb (1964)	0.135617

RMSE: 0.47240688235780776

MAE: 0.3957916413773053

PLOT:



2. SCENARIO 2: MATRIX FACTORIZATION USING NMF

OUTPUT:

Top 10 Recommended Movies for User 1

title

Star Wars: Episode V - The Empire Strikes Back (1980)	0.156916
Alien (1979)	0.145342
Raiders of the Lost Ark (Indiana Jones and the Raiders of the Lost Ark) (1981)	0.143347
Star Wars: Episode VI - Return of the Jedi (1983)	0.135678
Terminator, The (1984)	0.133093
Star Wars: Episode IV - A New Hope (1977)	0.132668
Back to the Future (1985)	0.132175
Wizard of Oz, The (1939)	0.129496
Casablanca (1942)	0.124046
Butch Cassidy and the Sundance Kid (1969)	0.122326

Name: 1, dtype: float64

RMSE: 0.49177321300974286

PLOT:

